

Language-Template Substitution for Structure-Compatible Generalization

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Phase 4: finite text-template generator transfer

Abstract

This phase tests structure-compatible generalization in a controlled finite text domain. Examples are rendered as short language templates with number words and offset words, while the deployment shift is precise: held-out number-word substitutions transport the label by the same offset. A finite generator is inferred from observed input/label-overlap evidence and used for OOD prediction and regularization without OOD labels.

1. Regime Transition

Earlier SCG phases tested oracle groups, inferred modular shifts, learned affine transports, and vision rotations. This phase keeps the finite auditability but moves the surface form into rendered language templates.

Field	Value
GPU	L4
configs	64
epochs	220
regularization	[0.0, 0.05, 0.2, 0.5]
max transforms	24
rows	258

Table 1. Language-template Modal L4 manifest.

2. Predictor Correlations

Domain	Predictor	Pearson r	Spearman r	N
language_template_substitution	compatibility_discovered	0.858	0.596	256
language_template_substitution	compatibility_true	0.853	0.628	256
language_template_substitution	negative_train_loss	0.492	0.232	256
language_template_substitution	id_validation_accuracy	0.478	0.127	256
language_template_substitution	train_accuracy	0.478	0.127	256
language_template_substitution	compatibility_wrong	-0.088	0.059	256
language_template_substitution	negative_abs_sharpness	-0.111	0.063	256
language_template_substitution	negative_parameter_l2	-0.476	-0.189	256

Table 2. Language-template predictors against OOD.

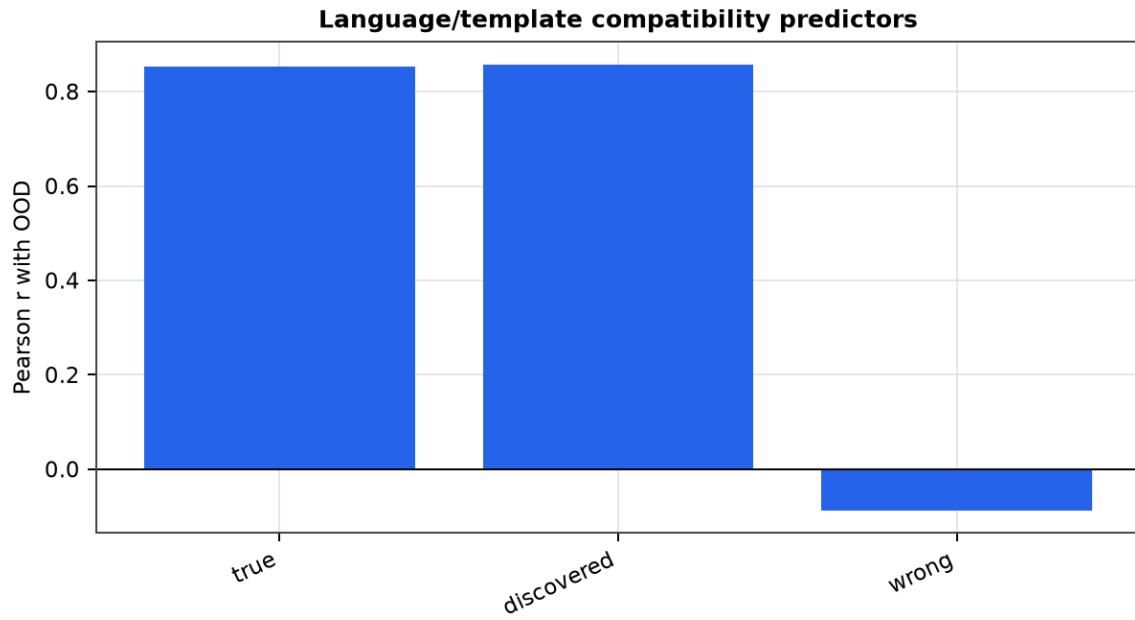


Figure 1. Language-template compatibility predictors.

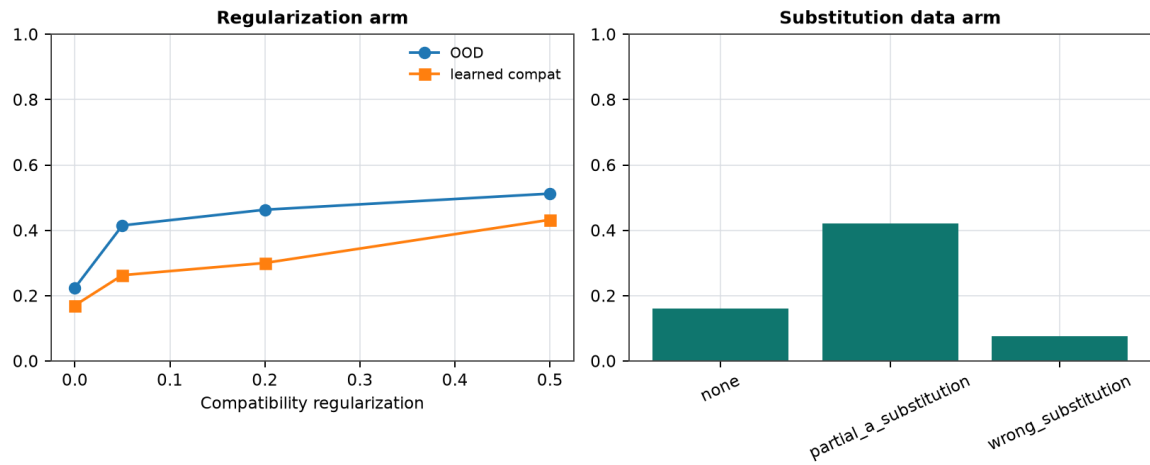


Figure 2. Language-template interventions.

3. Interventions

Reg	N	Mean train	Mean ID	Mean OOD	High-ID N	High-ID OOD
0.000	64	0.554	0.554	0.147	29	0.224
0.050	64	0.553	0.553	0.234	29	0.415
0.200	64	0.555	0.555	0.255	29	0.463
0.500	64	0.558	0.558	0.277	29	0.512

Table 3. Learned-substitution regularization arms.

Augmentation	N	Mean train	Mean OOD	Compat
none	68	0.569	0.161	0.152
partial_a_substitution	96	0.643	0.421	0.259
wrong_substitution	92	0.454	0.076	0.000

Table 4. Data-substitution arms.

4. Scope

The result is a controlled finite-language transfer test. It is not a claim about arbitrary natural-language paraphrase or semantic equivalence. The generator family is discovered inside an auditable template domain and evaluated on held-out number-word substitutions.